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Test - CO1 - AT3 CAS1417

1. Autonomous Vaccum - cleaning Robot.

a) PEAS Description

PEAS stands for performance Measure, Environment, Actuators, sensors. It describes the task environment of an intelligent agent.

Component	Description
Performance Measure	clean both rooms, minimize cleaning time and energy consumption, and avoid unnecessary movements.
Environment	Two rooms each of which can be clean or dirty.
Actuators	wheels or motors for moving Left and right a suction mechanism for cleaning and a control mechanism for doing nothing.
sensors	Dirt sensors to detect dirt and locations sensors to identify the robots room.

b) Types of intelligent Agent.

The most suitable agent is a Model-Based Relax Agent.

Justifications

1. It maintains or uses sensors information to detect dirt and determine its locations
2. If the current room is dirty the robot performs the suck action.

3. If the current room is clean, it moves to the other room

4. It can remember information about the other room even when that room is not visible.

Example :-

- If Room A is dirty → suck.
- If Room A is clean and Room B is dirty → Move Right
- If Both are clean → Do nothing.

2. Properties of Task Environments

The following table compares chess, autonomous driving and Puzzle solving.

Property	chess against a human	self-driving car	crossword puzzle
observability	Fully observable - it the chessboard and pieces visible	Partialy observable sensors cannot every road condition	Fully observable - the clues and grid are available.
Determini -stic	legal moves have predictible results, although the opponent's	other drivers Pedestrians, and road conditions	Filling a letter Produces a predictable result.
Sequential	each move effects future moves & results	each driving action detects decision & safety	each answer can constrain the answer
Dynamic	the board does not change a player	traffic and road conditions Changes	the grid and clues normally remains unchan
continuous	moves, Pieces and board Positions are distinct	speed, Position, steering, and time vary continuously	letters, cells, and word Placements are discrete.

Explanation :-

1. Chess :- chess is a fully observable, deterministic, sequential, static and discrete environment. Both players can see the board, and every legal moves has a defined.

2. self-driving car : Driving is partially observable, stochastic, sequential, dynamic, and largely continuous. The car must continuously sense its surroundings and react to chaning traffic conditions.

3. Crossword Puzzle : A crossword is generally fully observed deterministic, sequential, static and discrete.

Conclusion :- The properties of an environment influence the design of an intelligent and the algorithm.

3. Formulation of a Delivery Robot a Search Problem

The problem can be formulated using five components

i) Initial state.

The initial state is the starting Position of robot

$$S = (1, 1)$$

This means the robot starts in row 1, coloum 1 of grid.

ii) Goal state / Goal Test

The goal state is delivery Location.

$$G_1 = (4, 4)$$

$$\text{Goal Test}(S) = \begin{cases} \text{True}, & S = G_1 \\ \text{False}, & S \neq G_1 \end{cases}$$

iii) Actions

The robot has four possible actions.

- UP: Move one cell upward.
- Down: Move one cell downward
- Left: Move one cell to the left.
- Right: Move one cell to the right.

iv) Transition Model

The transition Model describes the result of performing an action is given state.

$$\text{Result}((r,c), \text{Right}) = (r, c+1)$$

This action moves the robot one cell to the right, Provided the destination is valid

v) Path cost Function

The Path cost is the total cost of all action taken from the starting state. to the goal state.

$$\text{Path cost} = \sum_{i=1}^{n} 1 = n$$

Example:-

S		X	
	X		
			X
X			G1

4. Breadth First Search

BFS explores nodes level by level. It uses a FIFO queue

Given edges.

S – A, S – B

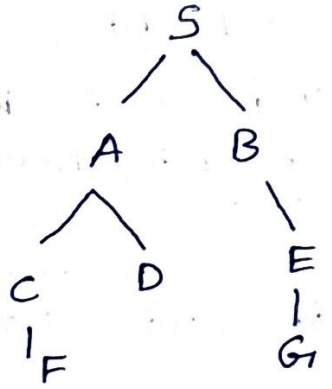
A – C, A – D

B – E

C – F

E – G1

search Tree



step by step

Node explored	S	A	B	C	D	E	F	G1
queue after expansion	A,B	B,C,D	C,D,E	E,F	E,G	F,G1	G1	Goal

Path cost = 3

BFS Properties

Let b be the branching factor and d be the depth

Property	Description
Time complexity	$O(b^{d+1})$
space complexity	$O(b^{d+1})$
completeness	complete when the branching
optimality	optimal when all step cost equ

5. Depth - First Search

DFS explores one branch as deeply as possible before backing. it uses a LIFO stack.

step by - step DFS.

Node explored	S	A	BC	F	D	B	E	G,
stack after	B, A	B, A, C	B, D, F	B, D	B	E	G,	goal.

Final Path and cost

S, A, C, F, D, B, E, G,

S → B → E → G,

Path cost = 3

Which Algorith is more suibable.

Reasons

1. BFS explores nodes in increasing order depth
2. It guarantees finding a goal if one exists
3. It finds the short Path when every edge